Name
Date
Course

Inductive and Deductive Reasoning (Homework)

In the following problems, make a conjecture based on the information given.

① For the last three weeks, Matt has looked up at the stars with his telescope on Saturday night.

Every Saturday night Matt looks up at the stars.

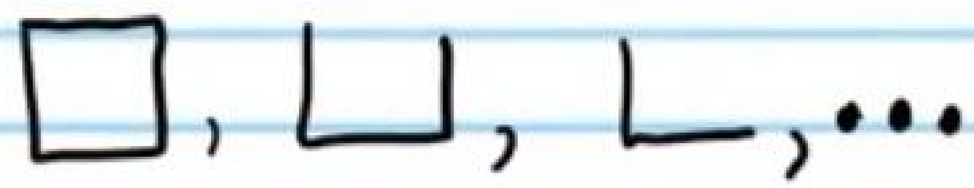
② Farmer Bob has three sterile mules.

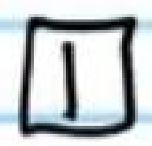
All mules are sterile

③ The price of gold has risen in the last three years.

The price of gold always rises.

Make a conjecture about the next thing in the sequence.

④ 



⑤ 19, 15, 11, . . .

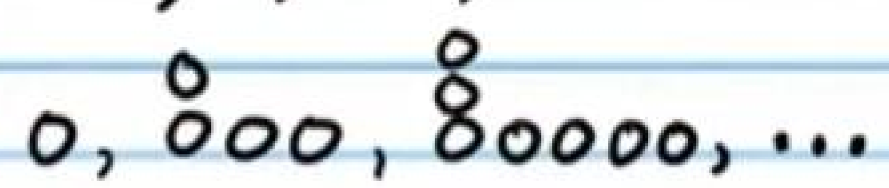
7

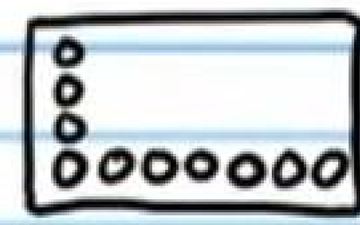
⑥ 125, 25, 5, . . .

1

⑦ −3, 9, −27, . . .

81

⑧ 



Complete the following conjectures based on the given information

⑨ I) $4 \times 6 = 24$
II) $2 \times 8 = 16$
III) $10 \times 4 = 40$

The product of two even integers is _____.

Even

⑩ I) $5 + 3 = 8$
II) $3 + 7 = 10$
III) $9 + 7 = 16$

The sum of two odd integers is _____.

Even

Find a counterexample for the following conjectures.

⑪ All apples are red.

Some apples are green.

⑫ Candy is always made of chocolate.

Jelly beans have no chocolate.

⑬ Every fish lives in the ocean.

Some fish live in lakes.

⑭ $(a+b)^2 = a^2 + b^2$ if a and b are real numbers.

There are many counterexamples.

$(1+2)^2 \neq 1^2 + 2^2$

$3^2 \neq 1 + 4$

$9 \neq 5$ ✓

Write the following statements in if-then form.

15. Linda cleaned the house because her friends arranged a visit.

If Linda's friends plan to visit, then she will clean the house.

16. The measure of an obtuse angle is between 90° and 180°.

If an angle is obtuse, then its measure is between 90° and 180°.

17. $x=9$ because $\sqrt{x}+5=8$.

If $\sqrt{x}+5=8$, then $x=9$.

18. All mammals have hair.

If an animal is a mammal, then it has hair.

19. Linear pairs are always supplementary angles.

If two angles form a linear pair, then they are supplementary.

a) Find the converse, inverse, and contrapositive of the statements below.
b) Find the truth value of each statement. If the statement is false, give a counterexample.
c) Is the statement biconditional? If so, write it in "if and only if" form.

20. If a polygon is a parallelogram, then it is a quadrilateral.

a) Converse: If a polygon is a quadrilateral, then it is a parallelogram.

Inverse: If a polygon is not a parallelogram, then it is not a quadrilateral.

Contrapositive: If a polygon is not a quadrilateral, then it is not a parallelogram.

b) False: A trapezoid is a quadrilateral, but not a parallelogram.

False: A trapezoid is not a parallelogram, but it is a quadrilateral.

True

c) The original statement is not biconditional because the original statement is true, but the converse is false.

21. If the measure of an angle is less than 90°, then it is an acute angle.

a) Converse: If an angle is acute, then its measure is less than 90°.

Inverse: If the measure of an angle is not less than 90°, then it is not an acute angle.

Contrapositive: If an angle is not acute, then its measure is not less than 90°.

b) True

True

True

c) Biconditional because the original and converse are true.

The measure of an angle is less than 90° if and only if it is an acute angle.

(22) If an integer is prime, then it is not divisible by two (note: this original statement is false).

a) Converse: If an integer is not divisible by two, then it is prime.
Inverse: If an integer is not prime, then it is divisible by two.
Contrapositive: If an integer is divisible by two, then it is not prime.

b) False: 9 is not divisible by two, but it is not prime.
False: 9 is not prime, but it's not divisible by two.
False: 2 is divisible by 2, but 2 is prime.

c) Not biconditional because the original and converse are false.

(23) Write the original statement in problem #22 with the symbols p and q. Then, use these symbols to write the converse, inverse, and contrapositive.

$p \to q$
$q \to p$
$\sim p \to \sim q$
$\sim q \to \sim p$

Determine if the Law of Detachment or the Law of Syllogism can be used with the following statements. Make a valid conclusion using one of the laws.

(24) If Jesse meets with his clients today, then he will leave his office. If he leaves his office, then he will eat at a restaurant.

Law of Syllogism
If Jesse meets with his clients today then he will eat at a restaurant.

(25) If a pet is a frog, then it is an amphibian. Lisa's pet is a frog.

Law of Detachment
Lisa's frog is an amphibian.

(26) If $x > 3$, then $x > 0$. If $x > 0$, then its a positive number.

Law of Syllogism
If $x > 3$, then x is a positive number.

(27) Tom liked the movie. If he liked the movie, then he will see it again.

Law of Detachment.
Tom will see the movie again.